The fastest particle passes at . By , greatest entry speed is times least ; their circular radii and in the field obey

The circle for speed passes origin , tangent to , and passes , giving

Center is at . The slower orbit has radius

lies at ; its circle is

Draw a tangent from , touching at . If it exits the field at with , it follows to . must lie on the field boundary.

Find . is normal to . Drop onto at ; drop onto at . Triangles and are similar, so

i.e.

giving

Completing the square:

Point at coordinates satisfies ,

Using , the coordinate of is

Thus shift boundary down parallel through to as shown. At it exits at along the circle tangent to reach .

6.

As move right, and first meet and collide. Impact is brief: both springs retain their natural lengths. are unchanged; have equal mass. Elastic impact leaves at rest and at , rightward. Just after impact, has rightward; rests. Mass center of and is the spring midpoint, at rightward; has rightward, rests. For and , mass center is also the spring midpoint, at rightward.

Origin follows the mass center rightward at . Its axis joins all four centers (mass-center frame). Take the first and impact as the time